

知识表示和推理作业

i) $\delta_0 = \varepsilon$, ~~$\{P(a, x, h(g(z))), P(z, h(y), h(y))\}$~~

ii) 差异集为 $D_0 = \{a, z\}$

iii) $\delta_1 = \delta_0 \setminus \{a/z\} = \{a/z\}$

$$\cancel{P(a, x, h(g(z)))} P(a, x, h(g(z))) \setminus \{a/z\} = P(a, x, h(g(a)))$$

$$P(z, h(y), h(y)) \setminus \{a/z\} = P(a, h(y), h(y))$$

iv) 差异集为 $D_1 = \{h(y), h(y)\}$

v) $\delta_2 = \delta_1 \setminus \{h(y)/h(y)\} = \delta_0 \setminus \{a/z\} = \{a/z, g(a)/y\}$

$$P(a, x, h(g(a))) \setminus \{g(a)/y\} = P(a, x, h(g(a)))$$

$$P(a, h(y), h(y)) \setminus \{g(a)/y\} = P(a, h(g(a)), h(g(a)))$$

vi) 差异集为 $D_2 = \{x, h(g(a))\}$

$$\delta_3 = \delta_2 \setminus \{x, h(g(a))\} = \{a/z, g(a)/y, h(g(a))/x\}$$

$$P(a, x, h(g(a))) \setminus \{h(g(a))/x\} = P(a, h(g(a)), h(g(a)))$$

$$P(a, h(g(a)), h(g(a))) \setminus \{h(g(a))/x\} = P(a, h(g(a)), h(g(a)))$$

最后一项为 $\delta_3 = \{a/z, g(a)/y, h(g(a))/x\}$

(2) 有理数都是实数: $A: \forall x (Q(x) \rightarrow R(x))$

无理数都是实数: $B: \forall x (W(x) \rightarrow R(x))$

有些数不是实数: $A_1: \exists x (\neg R(x))$

虚数既不是有理数也不是无理数: $A_2: \exists x (\neg Q(x) \wedge \neg W(x))$

$$A_1 = \forall x (\neg Q(x) \vee \neg R(x))$$

$$A_2 = \forall x (W(x) \vee R(x))$$

$$A_3 = \neg R(a)$$

$$\neg B = \forall x (Q(x) \vee W(x))$$

$$\text{约束 } S = \{\neg Q(x) \vee R(x), \neg W(x) \vee R(x), \neg R(a), \forall x (Q(x) \vee W(x))\}$$

$\Rightarrow$

$$① \neg Q(a) \vee R(a)$$

$$② \neg R(a)$$

$$③ \neg Q(a)$$

由①②得

$$④ Q(a) \vee W(a)$$

$$⑤ W(a)$$

由④⑤得

$$⑥ \neg W(a) \vee R(a)$$

$$⑦ R(a)$$

由⑥⑦得

$$⑧ \square$$

由⑧得

即证明 $A_1 \wedge A_2 \wedge A_3 \wedge B$ 不可满足

证明 $A_1, A_2, A_3 \rightarrow B$ 成立.

$$(3) \text{ 又又ANS: } \neg \neg \neg (Student(x) \wedge Happy(x) \vee Answer(x))$$

$$\text{子集 } S = \{ Student(john), Student(jane), StudentHappy(john) \vee Happy(jane),$$

$$\neg Student(john) \wedge Happy(jane) \vee Answer(jane) \}$$

$$① Student(john)$$

$$② \neg Student(john) \vee \neg Happy(jane) \vee Answer(jane)$$

$$③ \neg Happy(jane) \vee Answer(jane)$$

$$(3) Happy(john) \vee Happy(jane)$$

$$\text{分解为 } (1) Happy(john)$$

$$(2) Happy(jane)$$

$$① Student(john)$$

$$① Student(jane)$$

$$② Happy(john)$$

$$② Happy(jane)$$

$$③ Student(john) \wedge Happy(john) \text{ 成立}$$

$$③ Student(jane) \wedge Happy(jane)$$

$$\text{则 } \exists x (Student(x) \wedge Happy(x)) \text{ 且 } x \in John \vee Jane$$

即结果要么为 John, 要么为 Jane, 要么两者都有可能, 即存在 x 使 $Student(x) \wedge Happy(x)$ 成立